DATA SCIENCE

MODULE 5

DIMENSIONALITY REDUCTION

Study Notes • Concepts • PCA & SVD • Examples • Exam Questions

Learning objectives

• Understand the curse of dimensionality and why dimensionality reduction is useful.

• Differentiate feature selection from feature extraction.

• Explain Principal Component Analysis (PCA) and Singular Value Decomposition (SVD).

• Understand variance, covariance, eigenvectors, and eigenvalues in PCA.

• Learn other reduction techniques and practical implementation precautions.

Beginner-friendly classroom notes and examination revision material.

1. Introduction to Dimensionality

Definition: Dimensionality refers to the number of features or variables used to describe each data observation. A dataset with 100 input columns has 100 dimensions in its feature space.

Dimensionality reduction is the process of reducing the number of input dimensions while trying to preserve the most useful information in the data.

Why reduce dimensionality?

• **Reduce computation:** fewer variables can make training and prediction faster.

• **Reduce noise and redundancy:** some features may repeat information or contribute little useful signal.

• **Help visualization:** high-dimensional data can be projected to two or three dimensions for plotting.

• **Reduce overfitting risk:** fewer effective degrees of freedom may help models generalize.

• **Improve storage and efficiency:** compact representations may require less memory.

The curse of dimensionality

As the number of dimensions grows, data points become sparse in the feature space. Distance and neighborhood measures may become less informative, and much more data can be needed to learn reliable patterns. This collection of difficulties is called the curse of dimensionality.

Example: a model trained with 3 useful features may be easier to fit with limited data than one using hundreds of noisy features, although the best choice depends on the task.

2. Feature Selection vs. Feature Extraction

Feature Selection	Feature Extraction
Keeps a subset of original features.	Transforms original features into a smaller set of new features.
Original meaning is often easier to explain.	New components may be combinations of many original features.
Example: keep Age, Income, and Score from 30 columns.	Example: PCA transforms 30 columns into 5 principal components.

Feature selection chooses which columns to keep; dimensionality-reduction methods such as PCA often create new coordinates from the original columns.

3. Principal Component Analysis (PCA)

Definition: PCA is an unsupervised linear dimensionality-reduction technique that transforms correlated numeric variables into a smaller number of new, mutually orthogonal variables called principal components. The first component captures the greatest possible variance; each following component captures the greatest remaining variance subject to being orthogonal to earlier components.

Key terms

• **Variance:** measures how spread out values are.

• **Covariance:** indicates how two variables vary together.

• **Principal component:** a weighted linear combination of the original features.

• **Eigenvector:** a direction associated with a matrix transformation; in PCA it gives a principal direction.

• **Eigenvalue:** the amount of variance associated with a principal direction.

Steps of PCA

1. Arrange data as rows = observations and columns = features.
2. Clean missing values and usually standardize features when they have different units or scales.
3. Subtract each feature's mean (center the data).
4. Compute the covariance matrix, or use an equivalent decomposition of the centered data matrix.
5. Find eigenvalues and eigenvectors of the covariance matrix.
6. Sort components by decreasing eigenvalue.
7. Choose the first k components based on the desired information retention.
8. Project the centered data onto those k component directions.

Projection: if X is the centered data matrix and W contains the selected component directions, the reduced representation is Z = XW.

Why standardize features?

Suppose a dataset contains Age measured in years and Salary measured in thousands of rupees. Salary may have much larger numerical variance simply because of its scale. Standardization places features on comparable scales, often using z = (x – mean) / standard deviation. Whether to standardize depends on the data and task; it is generally important when units differ substantially.

Explained variance ratio

For component i, the explained variance ratio is its eigenvalue divided by the sum of all eigenvalues. The cumulative explained variance ratio adds these ratios from the first component through component k.

Example: if the first three components explain 50%, 30%, and 10% of the variance, retaining two components preserves 80% of the variance, while retaining three preserves 90%.

Advantages and limitations of PCA

• **Advantages:** reduces correlated dimensions, can compress data, may reduce noise, and supports 2D/3D visualization.

• **Limitations:** components may be hard to interpret; PCA is linear; high variance is not always the same as high predictive value; some information is lost when components are discarded.

4. Small Conceptual PCA Example

Imagine a dataset with two highly correlated measurements: height in centimeters and height in meters. They carry nearly the same information. PCA can find a direction that captures their shared variation and a second direction with very little variance. Keeping only the first component can represent most of the information with one dimension.

This example is conceptual: the actual PCA result depends on centering, scaling, and the values in the dataset.

5. Singular Value Decomposition (SVD)

Definition: Singular Value Decomposition factors a matrix A into three matrices: A = UΣV■. It is a general matrix factorization used in dimensionality reduction, least-squares problems, compression, and other numerical methods.

Meaning of the matrices

• **U:** contains left singular vectors.

• Σ **(Sigma):** diagonal matrix containing non-negative singular values, typically ordered from largest to smallest.

• **V■:** transpose of the matrix of right singular vectors.

A truncated SVD keeps only the largest k singular values and corresponding vectors to form a lower-rank approximation of the original matrix. This can retain dominant patterns while discarding smaller components.

Relationship to PCA: applying SVD to a centered data matrix provides principal directions and singular values from which PCA components and explained variance can be derived. SVD is often used to compute PCA numerically without explicitly forming the covariance matrix.

PCA vs. SVD

PCA	SVD
A statistical method that identifies directions of maximum variance.	A general factorization of a matrix.
Often explained using covariance, eigenvectors, and eigenvalues.	Expresses a matrix as UΣV■.
Usually centers the data; scaling depends on the task.	Can be applied to many matrices; for PCA, centering is normally required.

6. Other Dimensionality-Reduction Techniques

A. Linear Discriminant Analysis (LDA)

LDA is a supervised technique that finds projections intended to separate known classes. Unlike PCA, which does not use class labels, LDA uses labels and aims to increase separation between classes relative to variation within each class. In its standard form, the number of components is limited by the number of classes minus one.

B. t-SNE

t-distributed Stochastic Neighbor Embedding (t-SNE) is a nonlinear technique mainly used to visualize high-dimensional data in two or three dimensions. It aims to preserve local neighborhood relationships. Distances between separate clusters and cluster sizes in a t-SNE plot should not automatically be interpreted as global structure.

C. UMAP

Uniform Manifold Approximation and Projection (UMAP) is another nonlinear technique often used for visualization and exploratory analysis. It can preserve local structure and sometimes broader relationships, but results depend on settings and should be interpreted cautiously.

D. Autoencoders

An autoencoder is a neural network trained to reconstruct its input through a smaller internal representation called a latent code. The encoder maps input data to the compact code, and the decoder attempts to reconstruct the original input.

Technique	Supervision	Common purpose
PCA	Unsupervised	Linear compression and variance preservation.
LDA	Supervised	Project data to help separate labeled classes.
t-SNE	Usually unsupervised	Low-dimensional visualization of local neighborhoods.
UMAP	Usually unsupervised	Nonlinear embedding and visualization.
Autoencoder	Usually self-supervised reconstruction	Learn compact nonlinear representations.

7. Choosing the Number of Components

There is no single correct number of dimensions for every dataset. Common approaches include:

• **Explained variance threshold:** retain enough PCA components to preserve a chosen percentage, such as 90% or 95%, when appropriate.

• **Scree plot:** plot eigenvalues or explained variance and look for an elbow where additional components add less information.

• **Validation performance:** compare downstream model performance using different numbers of components.

• **Interpretability and cost:** balance compactness against information loss and computational requirements.

A variance threshold is a heuristic, not a guarantee of predictive performance. Components with low variance can sometimes contain information useful for predicting a target.

8. Practical Workflow and Common Mistakes

A safe general workflow is:

1. Split the data into training, validation, and test sets as appropriate.
2. Fit preprocessing steps, including imputation and scaling, using training data only.
3. Fit PCA or another reduction method on training data.
4. Transform validation and test data using the already-fitted preprocessing and reduction steps.
5. Select settings using validation or cross-validation.
6. Evaluate the final pipeline on held-out test data.

Common mistakes

• Fitting PCA on the entire dataset before splitting, which leaks information from test data.

• Forgetting to center data before standard PCA.

• Using PCA without considering scale differences among features.

• Assuming PCA uses target labels; standard PCA is unsupervised.

• Assuming more explained variance always means better classification or regression performance.

• Interpreting t-SNE or UMAP plots as exact maps of global distances.

9. Advantages and Limitations

Advantages	Limitations
Can reduce memory and computation.	Discarded dimensions may contain useful information.
Can remove redundancy and some noise.	Transformed features may be difficult to interpret.
Can make high-dimensional data easier to visualize.	Results depend on scaling, method, and parameters.
Can help some models generalize.	May not improve every model or dataset.

10. Short Exam Questions and Answers

1. Define dimensionality reduction.

Dimensionality reduction reduces the number of features or transforms data into fewer dimensions while preserving useful information.

2. What is the curse of dimensionality?

It describes problems caused by high-dimensional spaces, such as sparse data, less meaningful distances, and greater data requirements.

3. Differentiate feature selection and feature extraction.

Feature selection keeps some original features; feature extraction creates new transformed features.

4. Define PCA.

PCA is an unsupervised linear technique that projects data onto orthogonal directions of maximum variance.

5. What is a principal component?

A principal component is a linear combination of original features representing a direction in the transformed space.

6. What are eigenvalues in PCA?

They indicate the variance associated with the corresponding principal components.

7. Why is standardization sometimes necessary before PCA?

It prevents features with large numerical scales from dominating when their units or variances differ substantially.

8. What is explained variance ratio?

It is the fraction of total variance represented by a principal component.

9. Define SVD.

SVD factors a matrix A into UΣV■, where U and V contain singular vectors and Σ contains singular values.

10. State one relationship between PCA and SVD.

SVD of a centered data matrix can be used to compute PCA directions and explained variance.

11. Differentiate PCA and LDA.

PCA is unsupervised and maximizes variance; LDA uses class labels and seeks directions that separate classes.

12. What is t-SNE used for?

It is mainly used to visualize high-dimensional data in two or three dimensions while emphasizing local neighborhoods.

13. What is an autoencoder?

A neural network that learns a compact latent representation and reconstructs the input.

14. What is a scree plot?

A plot of component eigenvalues or explained variance used to help choose the number of components.

15. What is data leakage in dimensionality reduction?

It occurs when the reduction method is fitted using validation or test data before evaluation.

11. Long-Answer Practice

• Define dimensionality reduction and explain the curse of dimensionality.

• Explain PCA step by step, including centering, covariance, eigenvectors, eigenvalues, and projection.

• Explain explained variance ratio and discuss how to choose the number of principal components.

• Define SVD and explain the meaning of U, Sigma, and V■. Describe its relationship to PCA.

• Compare PCA, LDA, t-SNE, UMAP, and autoencoders.

• Discuss the advantages, limitations, and practical applications of dimensionality reduction.

• Describe a leakage-safe workflow for applying PCA in a machine-learning project.

Quick Revision

Term	Remember
Dimensionality reduction	Represent data using fewer dimensions.
Feature selection	Keep a subset of original columns.
PCA	Unsupervised linear projection; preserves directions of high variance.
Eigenvalue	Variance associated with a principal direction.
Explained variance	Fraction of total variance represented by selected components.
SVD	Matrix factorization A = UΣV■.
LDA	Supervised projection for class separation.
t-SNE / UMAP	Nonlinear embeddings commonly used for visualization.